



DESD – Embedded Linux Device Driver

1. Explain types of devices.
2. Explain difference between application programming and module programming.
3. Write a short note on debugging techniques.
4. Explain I/O architecture. Also explain three major hardware components of I/O architecture.
5. Which low level functions are used to access the device?
6. Explain Linux Device model (LDM/LDDM) in brief.
7. Explain arguments and return value of read() and write() system calls
8. What does struct cdev c_dev represent in kernel space?
9. How will you load and unload the module in kernel?
10. What is IOCTL, explain in detail?
11. What is recommended in device driver writing?
12. Explain features of Major/Minor number
13. Which function/KPI is used to copy from kernel space to user space and vice-a-versa.
14. Explain interrupt handling in Linux. Which are different types of bottom halves and where they are used?
15. What is significance of /proc.
16. How time is managed in kernel space?
17. What is difference between get_free_pages(), kmalloc() and vmalloc()? Explain mechanism behind them.
18. What is spinlock? How it differ from semaphore and mutex?
19. What is URB? Explain kernel functions for data transfer using and without URBs.